

Lecture 11 - Oct 5

Continuation from last lecture

$$\binom{\alpha}{k} = \begin{cases} \frac{\alpha(\alpha-1)\dots(\alpha-k+1)}{k!} & k \geq 1 \\ 1 & k = 0 \\ 0 & k < 0 \end{cases}$$

$$\boxed{\binom{\alpha}{k} \neq \binom{\alpha}{\alpha-k}}$$

The bottom must always be integer, and by definition $\alpha - k$ will not be an integer. This means the identity only holds if $\alpha \in \mathbb{Z}$.

5.4 - The multinomial Theorem

THEOREM (THE MULTINOMIAL THEOREM).

For positive integers $n, x_1, \dots, x_k$, where $t \geq 2$:

$$(x_1 + x_2 + \dots + x_t)^n = \sum \underbrace{\binom{n}{n_1, n_2, \dots, n_t}}_{\text{multinomial coeff.}} x_1^{n_1} x_2^{n_2} \dots x_t^{n_t}$$

where

$$\binom{n}{n_1, n_2, \dots, n_t} = \frac{n!}{n_1! n_2! \dots n_t!}$$

Proof. @ [Theorem 1 \(The multinomial Theorem\)](#).

$$(x_1 + \dots + x_t)^n = \underbrace{(x_1 + \dots + x_t) \dots (x_1 + \dots + x_t)}_{n \text{ factors}}$$

It can be seen that there are t^n terms in total (must choose some x_i for all n factors).

It can also be seen that the # of different terms is $\binom{n+t-1}{n}$, since it's the # of integral solutions to $n_1 + n_2 + \dots + n_t = n$ $n_1, n_2, \dots, n_t \geq 0$ (only the amounts of each x_i matter to find distinct terms).

So, let's group like terms for all t^n terms.

The coefficient of $x_1^{n_1}, x_2^{n_2}, \dots, x_t^{n_t} = \#$ ways to select n_1 copies of $x_1, \dots, n_t$ copies of x_t :

$$\begin{aligned} &= \binom{n}{n_1} * 1 * \binom{n-n_1}{n_2} * 1 * \dots * \binom{n-n_1-\dots-n_{t-1}}{n_t} * 1 \\ &= \boxed{\frac{n!}{n_1! n_2! \dots n_t!} = \binom{n}{n_1, n_2, \dots, n_t}} \end{aligned}$$

□

Examples

1

$(x_1 + \dots + x_5)^7$ is expanded

A. How many terms are there?

B. How many different terms are there?*

C. What is the coeff. of $x_1^2 x_2 x_4^3 x_5^*$

Solutions

There are 5^7 different terms.

There are $\binom{7+5-1}{5-1} = \binom{11}{4}$ different terms

Remark: there is no x_3 explicitly defined in the question, so it is implicit that the degree of x_3 is 0, so the total expression is $x_1^2 x_2^1 x_3^0 x_4^3 x_5^1$.

So, the coefficient is

$$\frac{7!}{2!1!3!1!} = \frac{7!}{2!3!}$$

2

$(2x_1 - 3x_2 + 5x_3)^6$

a. # terms

b. # of different terms

c. coeff of $x_1^3 x_2 x_3^2$

solution

terms: 3^6

different terms: $\binom{6+3-1}{3-1} = \boxed{\binom{8}{2}}$

For c:

$$\begin{aligned} & (2x_1)^3 (-3x_2)^1 (5x_3)^2 \\ &= \boxed{2^3 * (-3) * 5^2} \boxed{x_1^3 x_2 x_3^2} \end{aligned}$$

So,

$$\begin{aligned} \text{coeff. of } x_1^3 x_2 x_3^2 &= \frac{6!}{3!2!} * (2^3 (-3) 5^2) \\ &= \boxed{-6! * 2 * 5^2} \end{aligned}$$

Remark: The final coefficient will be the multinomial coefficient * the coefficient from expanding the coefficients of them individually

4.5 - Partially Ordered Sets

$a|b$: a is a divisor of b , or b is a divisible by a .

So, if we imagine the "operation" of divisibility on all integers, we can conclude the following:

$$\mathbb{Z}, | : \forall a, b \in \mathbb{Z}, a|b, b|a, a \nmid b, b \nmid a$$

In other words, for any integers a, b , we can see that at least one of the 4 conditions shown must occur

Given a set S , $\mathcal{P}(S)$ denotes the powerset of S .

$$\forall A, B \in \mathcal{P}(S), A \in B, B \in A, A \notin B, B \notin A$$

Basically given some "universe" (set in which elements exist), we can find some guarantees about those elements

DEFINITION (BINARY RELATION).

A **Binary Relation** on a set X is just a subset R of the cartesian product $X \times X = \{(x, y) | x \in X, y \in X\}$

DEFINITION (COMPARABLE AND INCOMPARABLE).

We say $a, b \in X$ are **comparable** if $(a, b) \in R$ or $(b, a) \in R$

We say $a, b \in X$ are **incomparable** if $(a, b) \notin R$ and $(b, a) \notin R$

We write $a R b$ if $(a, b) \in R$, or in other words, if a, b are comparable.

Example

let $X = \{1, 2, 3, 4, 5, 6\}$

Define R to be " $|$ " the divisibility among integers.

Or $R \subseteq X \times X$ with

$$R = \left\{ \begin{array}{l} (1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6) \\ (2, 2), (2, 4), (2, 6) \\ (3, 3), (3, 6), (4, 4), (5, 5), (6, 6) \end{array} \right\}$$

for example

$3 | 6 \iff 3$ and 6 are comparable

$3 \nmid 5 \iff 3$ and 6 are incomparable

A binary relation R on X is

- **reflexive** if $\forall x \in X, (x, x) \in R$
- **symmetric** if $(x, y) \in R \implies (y, x) \in R$
- **antisymmetric** if $(x, y) \in R$ and $(y, x) \in R \implies x = y$
- **transitive** if $(x, y) \in R$ and $(y, z) \in R \implies (x, z) \in R$

Example

"|" on $X = \{1, 2, 3, 4, 5, 6\}$

is reflexive (x divides x),

antisymmetric (if $a|b$ and $b|a$, then $a = b$),

and transitive (if $a|b$ and $b|c$, then $a|c$).

Another example:

$X = \mathcal{P}(S)$, " $\subseteq$ " relation on X

Then " $\subseteq$ " is reflexive, antisymmetric, and transitive

DEFINITION (PARTIAL ORDER).

A binary relation R on a set X is a **partial order** if it is reflexive, antisymmetric, and transitive.

Other definitions we won't cover right now:

Total order/linear order if it is reflexive, symmetric, and transitive.

DEFINITION (MINIMAL ELEMENT).

Let R be a binary relation on X . Then

$a \in X$ is a **minimal element** with respect to R if $(b, a) \in R$ implies that $b = a$ that is, there is no element $b \in X$ with $b \neq a$ such that $(b, a) \in R$

Examples

Minimal elements for divisibility relation: only 1 (only 1 divides 1, everything else gets divided by 1)

Minimal elements for subset relation: only $\emptyset$ (only $\emptyset \subseteq \emptyset$ and $\forall A \subseteq S, \emptyset \subseteq A$).

DEFINITION (MAXIMAL ELEMENT).

$a \in X$ is a **maximal element** with respect to $(a, b) \in R$ implies $b = a$, that is, there is not $b \in X$ such that $(a, b) \in R$

Maximal elements for divisibility relation: 4, 5, 6 (they don't divide anything else)

Maximal elements for subset relation: only whole subset S (it is only a subset of itself)